

2023/5/11 在义雨

1. ① 状态表示:

三元组 $\{m, c, b\}$

m 为左岸传道士人数

c 为左岸野人人数

b 为右岸传道士人数, 1 为左, 0 为右

② 算符 $L(a, b)$ 把 a 个传道士, b 个野人从左岸移到右岸
 $R(a, b)$ 把 a 个传道士, b 个野人从右岸移到左岸

③ 目标状态 $\{0, 0, 0\}$

④ 状态空间

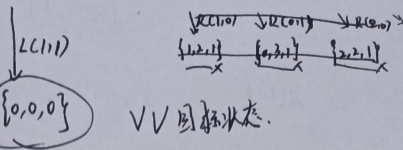
初始状态:

$\{3, 3, 1\}, \{4, 0, 1\}, \{4, 1, 0\}, \{2, 0, 2\}, \{2, 2, 0\}, \{2, 1, 1\},$

$\{0, 1, 1\}, \{0, 2, 2\}, \{1, 2, 1\}, \{1, 1, 2\}, \{0, 0, 0\}$

操作符

目标状态:



2. BFS

搜索次数

open 表

close 表

1

A

2

B C D

A

3

C A D E F G H

AB

4

D E F G H

ABC

5

E F G H

ABCD

6

F G H

ABCDE

7

G H

ABCDEF

8

H

ABCD EFG

9

ABCD EFG

ABCD EFG H

DPS

搜索次数

open 表

close 表

1

A

2

B C D

A

3

E F G H C D

AB

4

F G H C D

ABE

5

G H C D

ABEF

6

H C D

AB EFG

7

CD

AB EFG

一致代价

搜索次数

open 表

close 表

1

A(0)

A(0)

2

B(0) C(0) D(0)

A(0)

3

C(0) D(0) E(0) F(0) G(0) H(0)

A(0) B(0)

4

D(0) E(0) F(0) G(0) H(0)

A(0) B(0) C(0)

5

E(0) F(0) G(0) H(0)

A(0) B(0) C(0) D(0)

6

F(0) G(0) H(0)

A(0) B(0) C(0) D(0) E(0)

7

G(0) H(0)

A(0) B(0) C(0) D(0) E(0) F(0)

8

H(0)

A(0) B(0) C(0) D(0) E(0) F(0) G(0)

9

A(0) B(0) C(0) D(0) E(0) F(0) G(0) H(0)

搜索过程

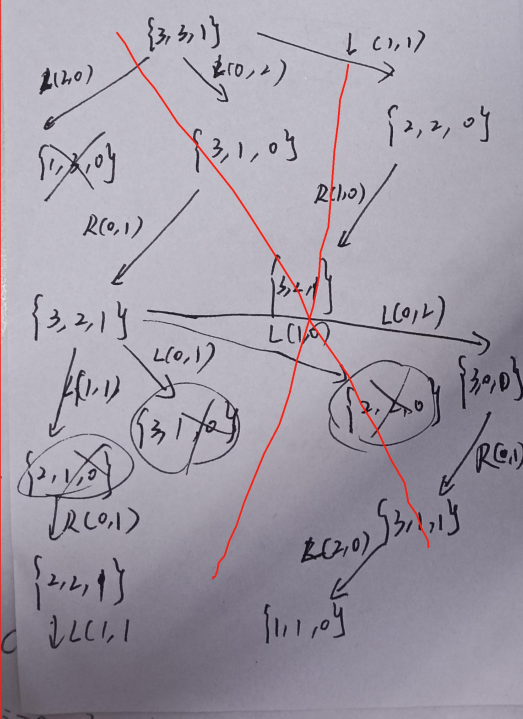


Diagram illustrating the search space for the 3-disk Tower of Hanoi problem, starting from the initial state $\{3, 3, 1\}$. The diagram shows various states and transitions, with many states marked as unreachable (X) or already visited (X).

Key states and transitions shown:

- Initial state: $\{3, 3, 1\}$
- Transitions from $\{3, 3, 1\}$:
 - $LC(0,0) \rightarrow \{1, 3, 0\}$ (X)
 - $LC(0,2) \rightarrow \{3, 1, 0\}$
 - $LC(1,1) \rightarrow \{2, 2, 0\}$
- Transitions from $\{3, 1, 0\}$:
 - $RC(0,1) \rightarrow \{3, 2, 1\}$
 - $RC(0,2) \rightarrow \{3, 3, 1\}$ (X)
- Transitions from $\{2, 2, 0\}$:
 - $RC(1,0) \rightarrow \{3, 2, 1\}$ (X)
 - $RC(1,1) \rightarrow \{2, 3, 1\}$ (X)
 - $RC(1,2) \rightarrow \{3, 3, 1\}$ (X)
- Transitions from $\{3, 2, 1\}$:
 - $LC(0,0) \rightarrow \{2, 2, 0\}$ (X)
 - $LC(0,1) \rightarrow \{3, 1, 0\}$ (X)
 - $LC(0,2) \rightarrow \{1, 2, 0\}$ (X)
 - $LC(1,1) \rightarrow \{2, 3, 1\}$ (X)
 - $LC(1,2) \rightarrow \{3, 0, 0\}$
 - $RC(0,1) \rightarrow \{2, 1, 0\}$ (X)
 - $RC(0,2) \rightarrow \{3, 1, 1\}$
- Transitions from $\{2, 3, 1\}$:
 - $LC(1,1) \rightarrow \{2, 2, 0\}$ (X)
 - $LC(1,2) \rightarrow \{3, 0, 0\}$ (X)
 - $RC(1,1) \rightarrow \{2, 1, 0\}$ (X)
 - $RC(1,2) \rightarrow \{3, 1, 1\}$ (X)
- Transitions from $\{3, 0, 0\}$:
 - $RC(1,0) \rightarrow \{2, 1, 0\}$ (X)
 - $RC(1,1) \rightarrow \{1, 2, 0\}$ (X)
 - $RC(1,2) \rightarrow \{3, 1, 1\}$ (X)
- Transitions from $\{2, 1, 0\}$:
 - $RC(1,0) \rightarrow \{1, 2, 1\}$ (X)
 - $RC(1,1) \rightarrow \{0, 3, 1\}$
 - $RC(1,2) \rightarrow \{2, 2, 1\}$ (X)
 - $RC(2,1) \rightarrow \{1, 3, 1\}$ (X)
- Transitions from $\{1, 2, 0\}$:
 - $RC(1,0) \rightarrow \{1, 2, 1\}$ (X)
 - $RC(1,1) \rightarrow \{0, 3, 1\}$
 - $RC(1,2) \rightarrow \{2, 2, 1\}$ (X)
 - $RC(2,1) \rightarrow \{1, 3, 1\}$ (X)
- Transitions from $\{0, 3, 1\}$:
 - $LC(0,2) \rightarrow \{0, 4, 0\}$
 - $LC(1,1) \rightarrow \{1, 1, 1\}$
 - $LC(1,2) \rightarrow \{0, 2, 1\}$
 - $RC(0,1) \rightarrow \{1, 1, 1\}$
 - $RC(0,2) \rightarrow \{0, 2, 1\}$
 - $RC(1,1) \rightarrow \{1, 2, 1\}$ (X)
 - $RC(1,2) \rightarrow \{0, 3, 1\}$
 - $RC(2,1) \rightarrow \{1, 3, 1\}$ (X)
 - $RC(2,2) \rightarrow \{0, 4, 0\}$
- Transitions from $\{1, 1, 1\}$:
 - $LC(1,1) \rightarrow \{0, 0, 0\}$ (Goal state)

The diagram concludes with the goal state $\{0, 0, 0\}$ circled, and the text "VV 目标状态" (VV Goal state) written below it.

2023/05/01 王文明

贪心
搜索次数

open表

close表

1	A(0)	
2	B(1) C(1) D(5)	A(0)
3	C(0) E(0) F(2) G(2) H(9) D(5)	A(0) B(3)
4	E(0) F(0) G(2) H(9) C(1)	A(0) B(3) D(5) A(0) B(3) D(5)
5	F(0) G(2) H(9) C(1)	A(0) B(3) D(5) E(1)
6	G(0) H(9) C(1)	A(0) B(3) D(5) E(1) F(0)
7	H(9) C(1)	A(0) B(3) D(5) E(1) F(0) G(0)
8	C(1)	A(0) B(3) D(5) E(1) F(0) G(0) H(9)

A* 评估函数 $f = g + h$ $g = \text{cost}(path)$ $h = 0$

搜索次数

open表

close表

1	A(0)	
2	B(3) D(5) C(1)	A(0)
3	H(0) B(3) D(5) E(1) F(0) G(0) C(1)	A(0) B(3)
4	D(5) E(1) F(0) G(0) C(1)	A(0) B(3) H(0)

B.

